Reproducibility in Computational Experiments

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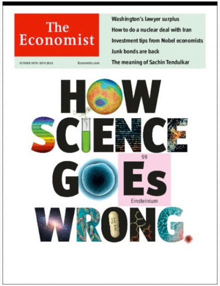
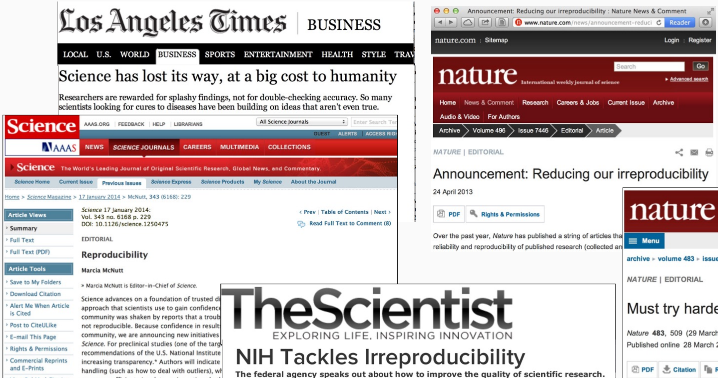
Motivation and Introduction

Public Evidence for a Lack of Reproducibility

- J.P. Ioannidis. *Why Most Published Research Findings Are False* PLoS Med. 2005.
- *Lies, Damned Lies, and Medical Science*, The Atlantic. Nov, 2010
- *Reproducibility: A tragedy of errors*, Nature, Feb 2016.
- Steen RG, *Retractions in the scientific literature: is the incidence of research fraud increasing?*, J. Med. Ethics 37, 2011





Courtesy V. Stodden, SC, 2015

Dong-Pyou Han Assistant professor, Biomedical sciences, Iowa State University, 2013

Falsified blood results to make it appear as though a vaccine exhibited anti-HIV activity

- Han and his team received $\approx \$19$ million from NIH
- 1 retracted publication and resignation of university. Sentenced in 2015 to 57 months imprisonment for fabricating and falsifying data in HIV vaccine trials He was also fined US $\$7.2$ million!

Newsworthy Stories about Scientific Misconduct

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Dieterik Stapel Professor, Social Psychology, Univ. Amsterdam, 2011

I failed as a scientist. I adapted research data and fabricated research. Not once, but several times, not for a short period, but over a longer period of time. [..] I am aware of the suffering and sorrow that I caused to my colleagues... I did not withstand the pressure to score, to publish, the pressure to get better in time. I wanted too much, too fast. In a system where there are few checks and balances, where people work alone, I took the wrong turn. *58 retracted publications*

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Brian Wansink Professor, Psychological Nutrition, Cornell, 2016

When she arrived, I gave her a data set of a self-funded, failed study which had null results. I said "This cost us a lot of time and our own money to collect. There's got to be something here we can salvage because it's a cool (rich & unique) data set." I told her what the analyses should be and what the tables should look like. [..] Every day she came back with puzzling new results, and every day we would scratch our heads, ask "Why," and come up with another way to reanalyze the data with yet another set of plausible hypotheses *17 retracted publications*

A Credibility Crisis?

Scientific misconduct is obviously wrong but it's not new!

- Every domain has its black sheep
- The publish or perish pressure is a huge pain

Media attention inflates conspiracy opinions ☹

Scientific result are worthless. Stop the scientific dictatorship/lobby!

The Battle against Scientific Fraud

CNRS International Magazine




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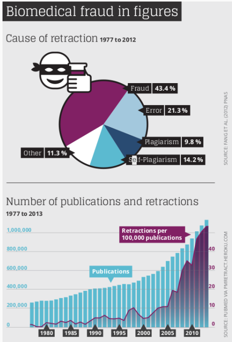


Fraud is the (uninteresting) visible part of the iceberg

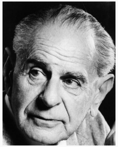
- **Failing to reproduce the results of others is common**

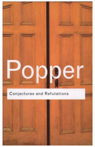
  1,500 scientists lift the lid on reproducibility, Nature, May 2016
- How so? Why now? Why is this important? What can we do about it?

Reproducibility of experimental results is the *hallmark of science*

1934: **Karl Popper** puts the notions of falsifiability and crucial experiment as the hallmark of science

- If no experiment can be set up to disprove your theory, it is not science
- Good experiments discriminate good theories from bad ones
- Non-reproducible single occurrences are of no significance to science





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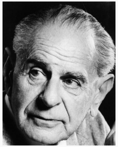
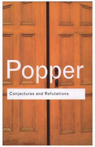

An ideal rather than the norm

Popper's proposal works well for Physics from the 18th century but is not so simple for many other domains:

- Theory of evolution
- Spotting a SuperNova
- Particle Physics (a single LHC)
- Biology (every animal does not behave in the same way)
- Anthropology (impact on people from a remote culture)

Reproducibility: a *core value of science*

1. Universality: Science aims for objective findings, accessible to anyone

   Reproducibility acts as a Universality/Robustness control

2. Incremental: We build on each others work but everybody makes mistakes

   Methods, biases, ... How to discriminate sound ~~theories~~ experiments from bad ones? 😉

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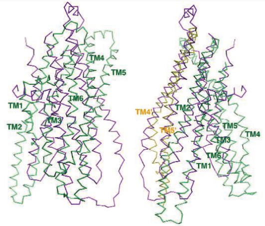
But, scientific practices have greatly evolved, in particular since we rely on computers

How computers broke science – and what we can do about it

– Ben Marwick, The conversation, 2015



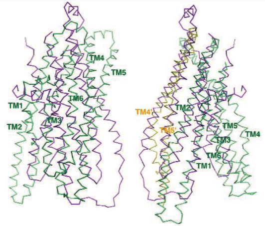
Geoffrey Chang (Scripps, UCSD) works on crystalography and studies the structure of cell membrane proteins.

He specialized in structures of multidrug resistant transporter proteins in bacteria: MsbA de Escheria Choli (Science, 2001), Vibrio cholera (Mol. Biology, 2003), Salmonella typhimurium (Science, 2005)

2006: Inconsistencies reveal a programming mistake

> *A homemade data-analysis program had flipped two columns of data, inverting the electron-density map from which his team had derived the protein structure.*

5 retractations that motivate improved software engineering practices in comp. biology



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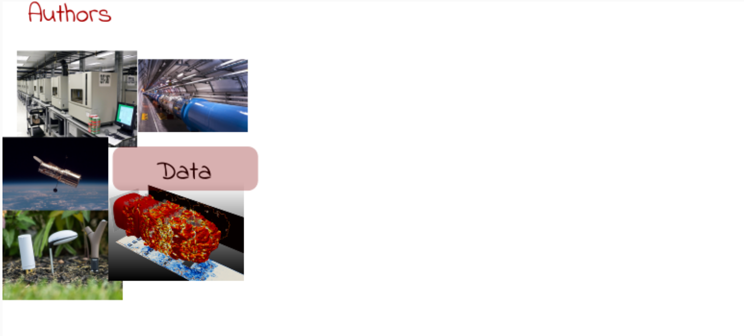
There is worse!

- The generalized and intensive use of spreadsheets (COVID tracing)
- Relying on black box statistical methods is infinitely easier than understanding them
  - Learning and Data Analytics frameworks are nuclear weapons
- Numerical errors and software environment unawareness

The processing steps between raw observations and findings have gotten increasingly numerous and complex.



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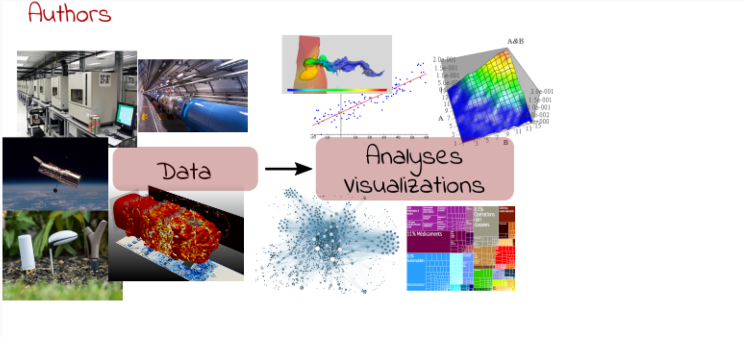

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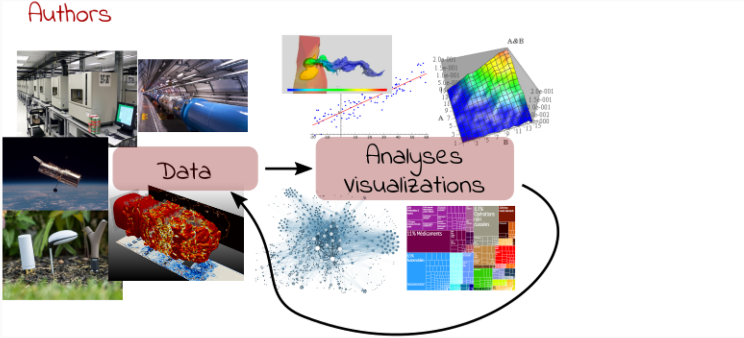

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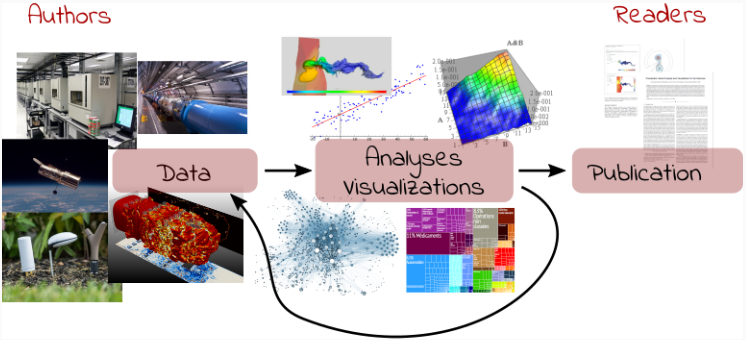

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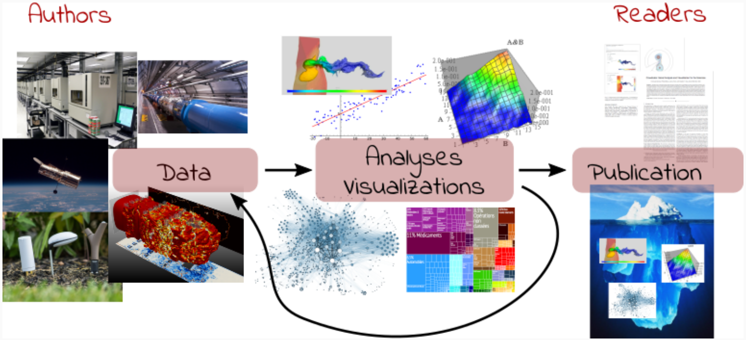

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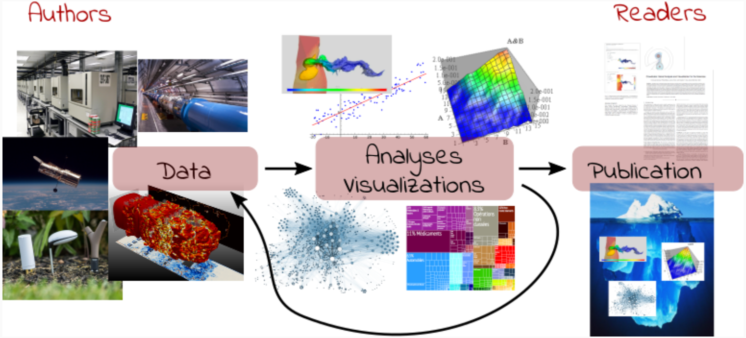
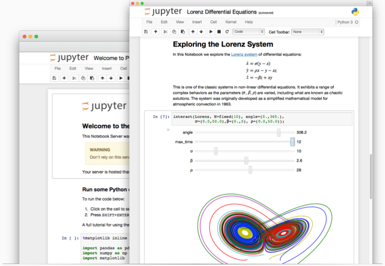


Reproducible Research = Bridging the Gap by working Transparently

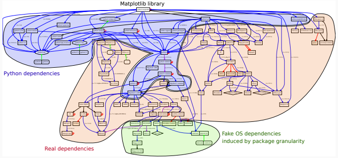
Tools

Existing Tools, Emerging Standards

Notebooks and workflows



Software environments



Sharing platforms

Tool 1: Computational Notebooks/Literate Programming

Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

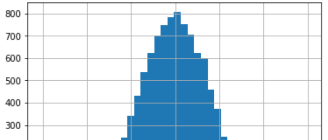
```
3.141592653589793
```

Mais calculé avec la **méthode** des [aiguilles de Buffon](), on obtiendrait comme **approximation** :

```
import numpy as np
N = 1000000
x = np.random.uniform(size=N, low=0, high=1)
theta = np.random.uniform(size=N, low=0, high=pi/2)
2/(sum((x+np.sin(theta))>1)/N)
```

```
3.1437198694098765
```

On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des *dessins qui n'ont rien à voir* avec π (si ce n'est une constante de normalisation... ☺).



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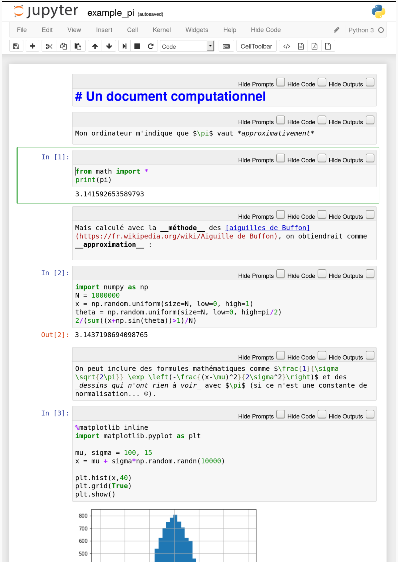

jupyter example_pi (autosaved)

File Edit View Insert Cell Kernel Widgets Help Hide Code Python 3

Code CellToolbar

Hide Prompts Hide Code Hide Outputs

Un document computationnel

Hide Prompts Hide Code Hide Outputs

```
Mon ordinateur m'indique que $\pi$ vaut *approximativement*
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In [1]: Hide Prompts Hide Code Hide Outputs

```
from math import *
print(pi)
```

```
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Hide Prompts Hide Code Hide Outputs

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In [2]: Hide Prompts Hide Code Hide Outputs

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Out[2]: 3.1437198694098765

Hide Prompts Hide Code Hide Outputs

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```

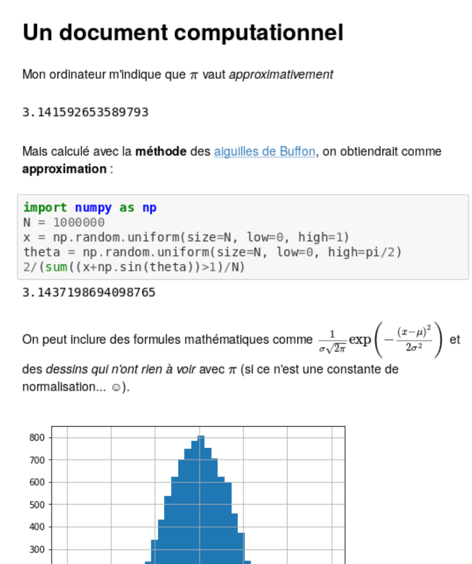
In [3]: Hide Prompts Hide Code Hide Outputs

```
%matplotlib inline
import matplotlib.pyplot as plt

mu, sigma = 100, 15
x = mu + sigma*np.random.randn(10000)

plt.hist(x,40)
plt.grid(True)
plt.show()
```

Document final



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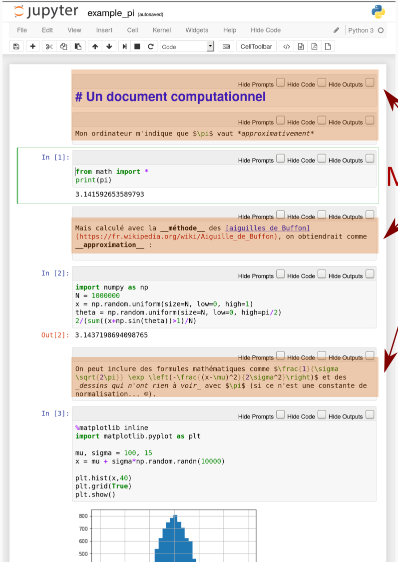

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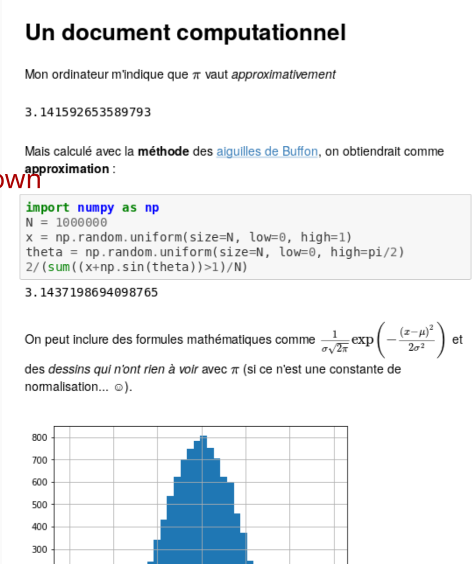

Tool 1: Computational Notebooks/Literate Programming

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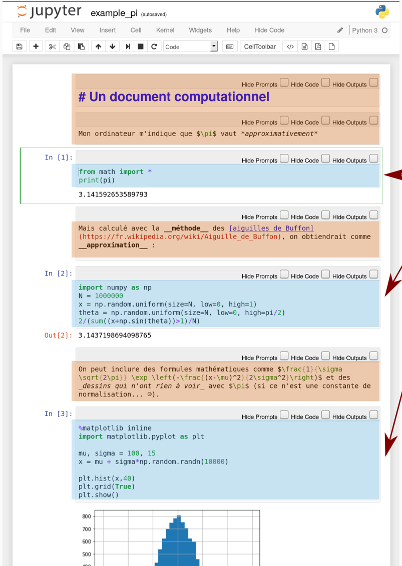

Code

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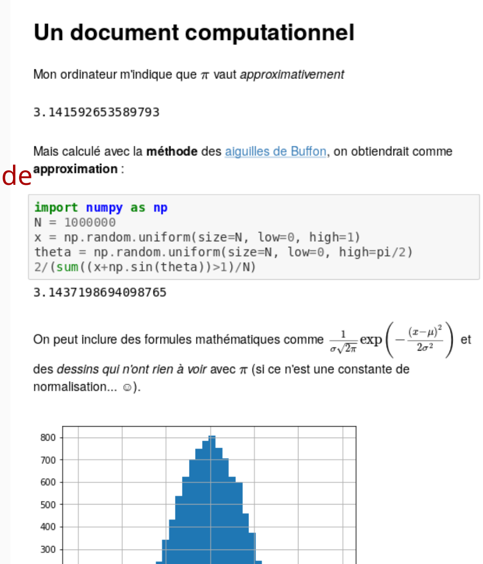

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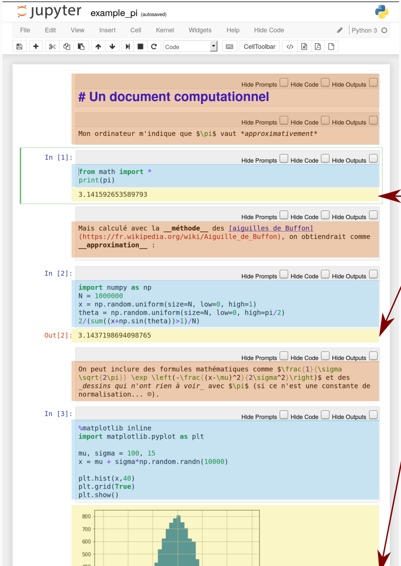

Résultats

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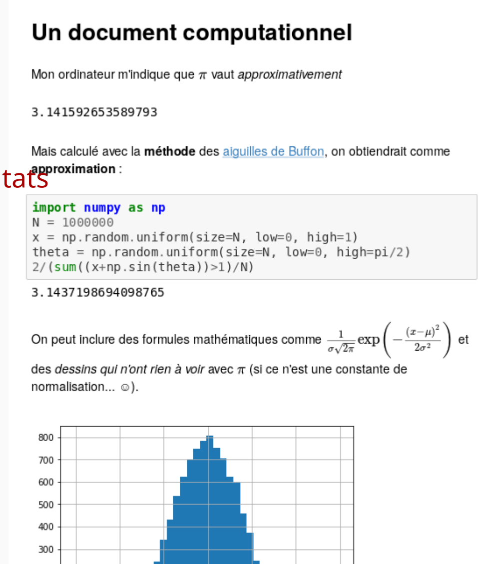

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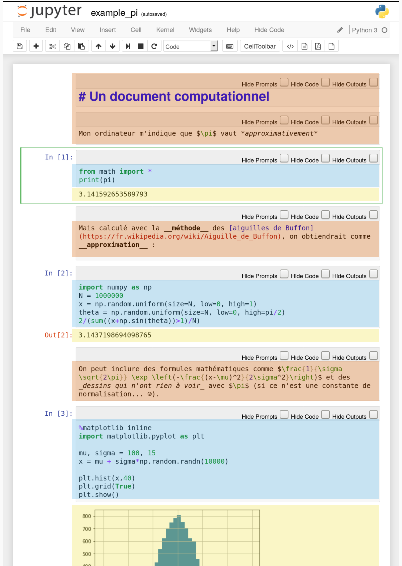

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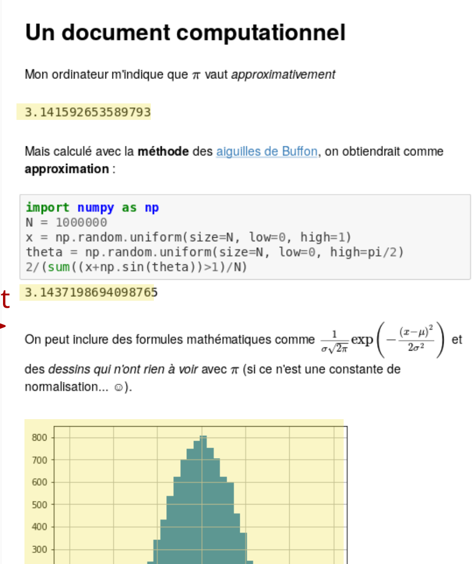

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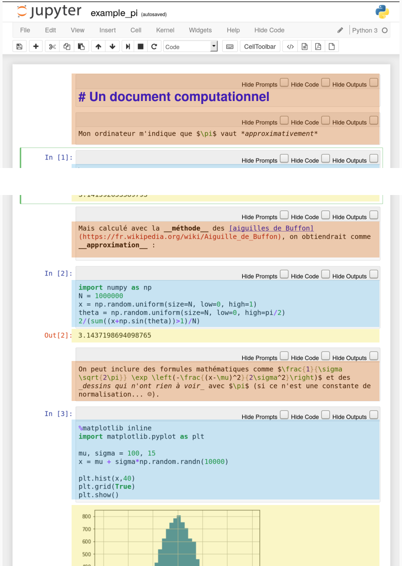

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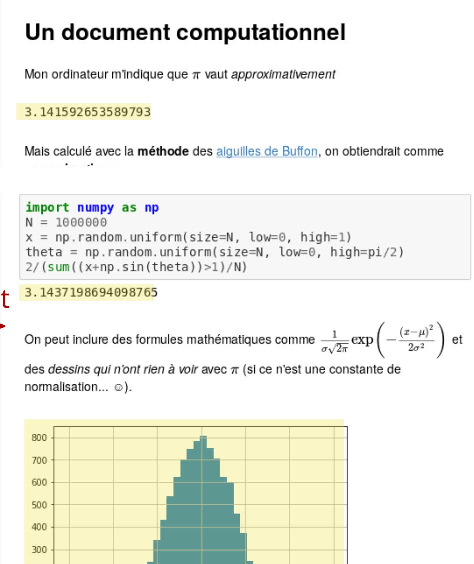

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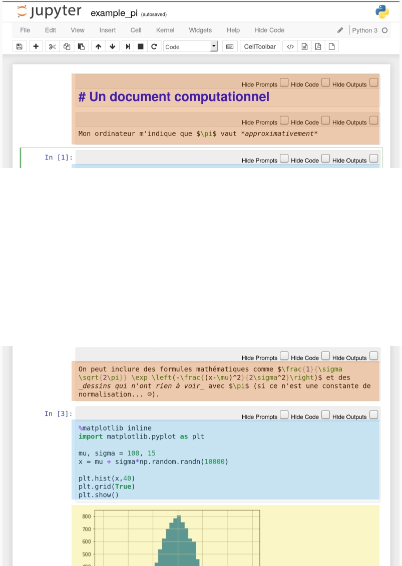

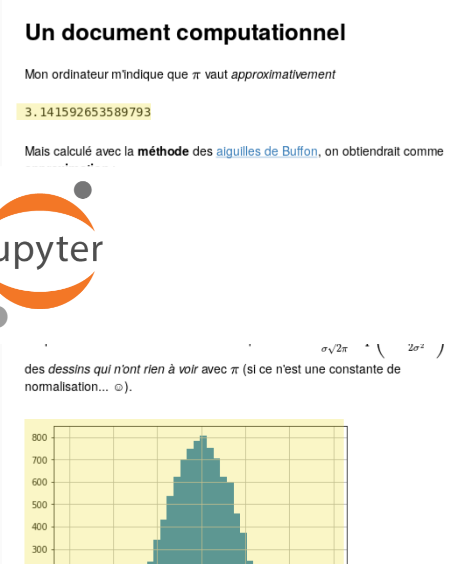

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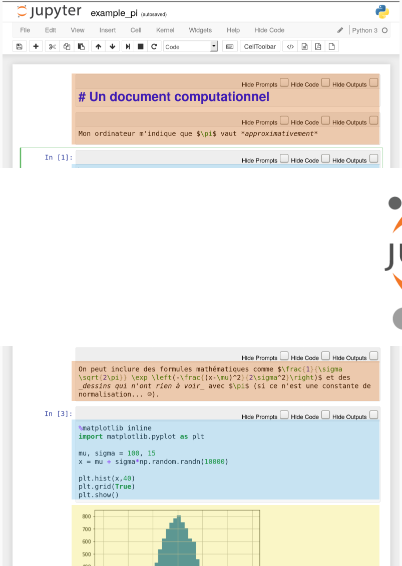

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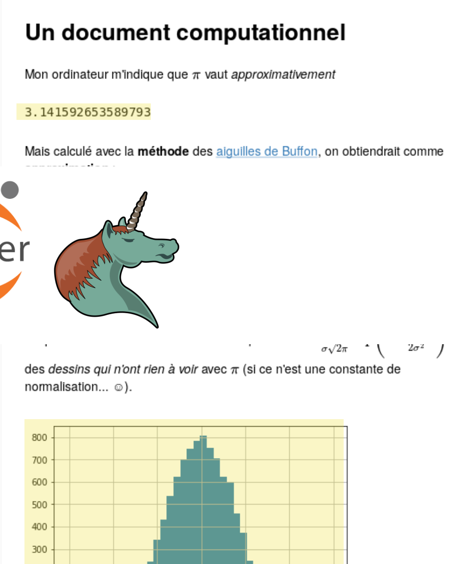

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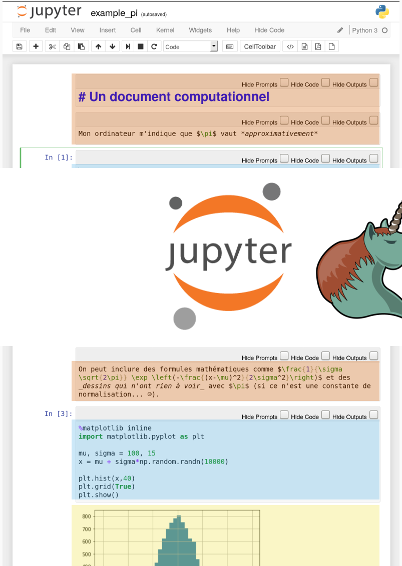

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Tool 2: Fighting Software Environments Nightmare

What is hiding behind a simple

```
Package: python3-matplotlib
Version: 2.1.1-2
Depends: python3-dateutil, python-matplotlib-data (>= 2.1.1-2),
python3-pyparsing (>= 1.5.6), python3-six (>= 1.10), python3-tz,
libjs-jquery, libjs-jquery-ui, python3-numpy (>= 1:1.13.1),
python3-numpy-abi9, python3 (<< 3.7), python3 (>= 3.6~),
python3-cycler (>= 0.10.0), python3:any (>= 3.3.2-2~), libc6 (>=
2.14), libfreetype6 (>= 2.2.1), libgcc1 (>= 1:3.0), libpng16-16 (>=
1.6.2-1), libstdc++6 (>= 5.2), zlib1g (>= 1:1.1.4)
```

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```
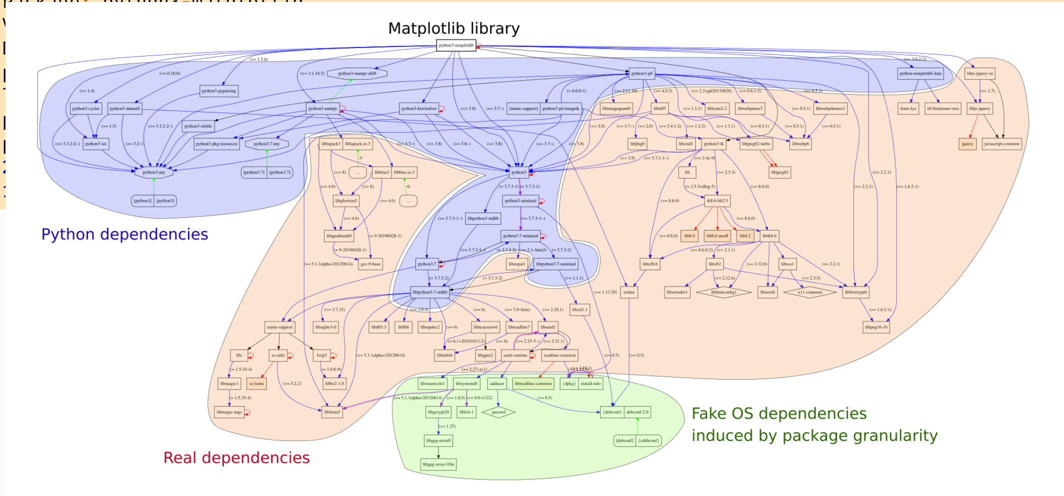



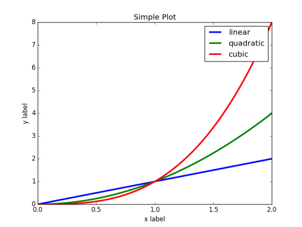
Tool 2: Fighting Software Environments Nightmare

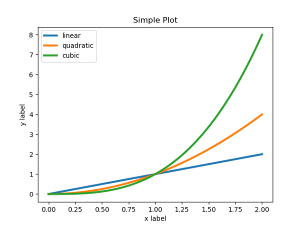
Python and its rapidly evolving environment

```
python2 -c "print(10/3)"
python3 -c "print(10/3)"
```

```
3
3.3333333333333335
```

Python and its rapidly evolving environment





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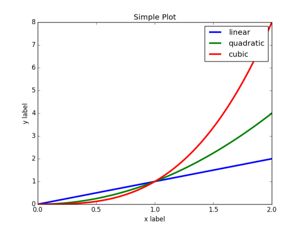
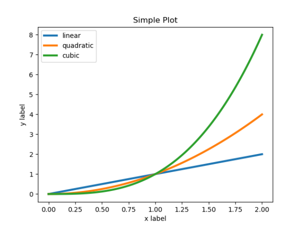
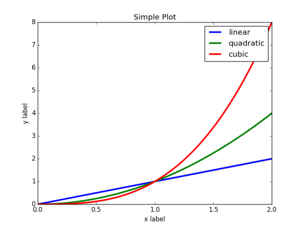
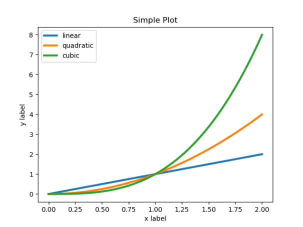




Cortical Thickness Measurements (PLOS ONE, June 2012): *FreeSurfer: differences were found between the Mac and HP workstations and between Mac OSX 10.5 and OSX 10.6.*

Python and its rapidly evolving environment





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Tool 3: Fighting Information Loss with Archives

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Article archives arXiv.org HAL archives-ouvertes.fr

Data archives figshare zenodo

Software Archive Software Heritage

or = awesome collaborations $\neq$ archive

Reproducibility

Different (but Converging) Reproducibility Concerns

Reproducibility/robustness of the scientific fact, the statistical analysis, the computation, the observation, the process, … ?

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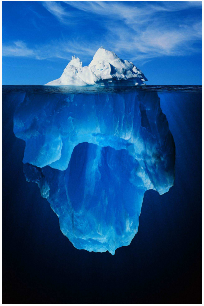

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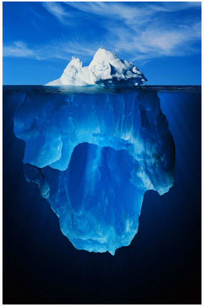

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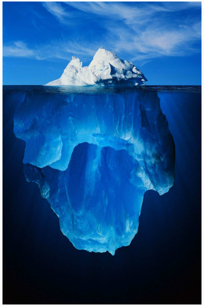

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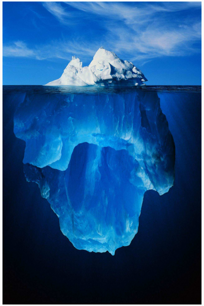

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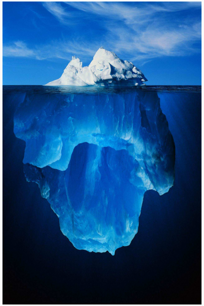
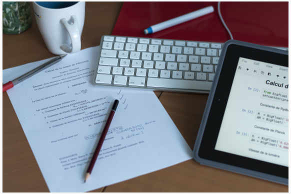


This requires first class software engineering practices instead of building on prototypes

Software factories, Archives, and Provenance Tracking tools

Closure

Soft. Engineering, Statistics, and Reproducible Research in the curricula



- Book on RR *Vers une recherche reproductible: Faire évoluer ses pratiques*
- MOOC on RR (3rd edition Feb. 2020)
- A new "Advanced RR" MOOC (1st edition 2024)
  - Managing research data (HDF5, archiving)
  - Software environment control (Docker, Guix)
  - Scientific workflow (snakemake)

Manifesto: *"I solemnly pledge"* (WSSSPE, Lorena Barba, FAIR)

1. I will teach my graduate students about reproducibility
2. All our research code (and writing) is under version control
3. We will always carry out verification and validation
4. We will share data, plotting script & figure under CC-BY
5. We will upload the preprint to arXiv at the time of submission of a paper
6. We will release code at the time of submission of a paper
7. We will add a "Reproducibility" declaration at the end of each paper
8. I will keep an up-to-date web presence

Changing Publishing Practices

Artifact evaluation and ACM badges

Major conferences

- Supercomputing: Artifact Description (AD) mandatory, Artifact Evaluation (AE) still optional, Double blind vs. RR
- NeurIPS, ICLR: open reviews, reproducibility challenge

  Joelle Pineau @ NeurIPS'18 
- ACM SIGMOD 2015-2019, Most Reproducible Paper Award...

Mentalities are evolving people care, make stuff available, errors are found and fixed

Resources and Acknowledgments

Resources and Acknowledgments

Slides are available here

https://github.com/schnorr/SMPE/raw/master/lectures/talk_26_03_20_SemIntegracaoPPGC.pdf

Thanks to

- Arnaud Legrand for sharing slides and knowledge about this topic
- My laboratory students that have understood the importance
  - It may be hard to face/learn new things, but they are essential to evolve!

MOOC Reproducible Research II: Practices and tools for managing computations and data, Learning Lab Inria

- Konrad Hinsen, Arnaud Legrand, Christophe Pouzat
- 3rd Edition: May 2026 – December 2026

Rede Brasileira de Reprodutibilidade (RBR), participe!

https://www.reprodutibilidade.org/